

Evolution and/or Creation Theory: A Science Teacher's Dilemma



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By Aaron Powner

Abstract: The purpose of this paper is to describe an issue in which society and public education interact with controversy. The call for evolution education in science classrooms has caused concerns and, in many instances, provoked anger and defensive retaliation by religious persons who believe in creationism. I discuss both sides of the controversy, legal considerations, education theory issues, and pedagogical considerations. It is my intent to determine in an academically justifiable manner the most appropriate way for me, as a science teacher, to approach the subject of evolution in my classroom.

Background of the Conflict: Man as a Truth Seeker

In the beginning there was religious tradition. The one thing that seems to have changed little over countless generations of humanity is our need to understand our surroundings, who we are, where we came from, and what lies ahead. Oral traditions, wisdom of the elders, spiritual journeys, or simple aesthetic awareness; all of these and more have been the predominant way to understand the human condition until relatively recent generations. Since the days before Aristotle, Plato and other ancient philosophers, mankind has been relying increasingly upon human intellect to observe and explain reality.

Though many of us yet retain a belief in higher powers, the development of science as a universal method of discovery and progress has become a fundamental part of our society. In general, society has viewed science and technological innovation as a welcome tool for bettering the human condition. With increasing frequency, however, developing philosophies and theories of science are conflicting with long-held traditions or beliefs that are precious to individual cultures. The evolution/creationism issue is a prime example of such conflict.

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In cases where science conflicts with public opinion or popularly held religious beliefs, both sides ask poignant questions. Science advocates ask, "Can public opinion determine what is true?" Defenders of religious faith ask, "Can human logic define ultimate truth?" The theory of evolution presents a controversy that touches the foundations of our understandings of reality. Many students and parents are either openly or secretly hostile towards evolution education. Many see it as a designed attack on their upbringing and religious heritage (Clough 1994). Consequently, over the past several decades, this area of science education has become a fiery political issue. How does a science teacher determine when, how, and whether at all to teach the theory of evolution?

In this political and philosophical debate, it is enlightening to evaluate what Americans actually believe. Gallup polls from November 1991 reported that 47% of those polled, including a fourth of the college graduates, agreed with the statement that, "God created man pretty much in his present form at one time within the last 10,000 years." A further 40% agreed with the more moderate statement, "Man has developed over millions of years from less advanced forms of life, but God guided this process, including man's creation." Additionally, 58% of the American public favors teaching creationism in schools. Only 9% agreed that natural processes now operating have led to the advent of humans over time, without the benefit of some sort of conscious, active, ongoing, divine intervention (Firenze 1997, Jackson 1995).

These polls indicate that 87% of the citizens who make up our nation believe in God or some form of intelligent universal power. Most Americans are at least nominally theistic and a substantial percentage of us build our lives on theistic principles.

In light of the varied beliefs and values of our nation, what should be the role of our public schools? Is the point of public education to indoctrinate our children with our common cultural and societal values or to broaden their horizons – increasing their intellectual freedom? In the latter case, science teachers have often been accused of stepping across their disciplinary limits, indirectly or directly suggesting that students reevaluate their religious beliefs to accommodate scientific ideas (Dagher 1997).

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Some teachers shy away from teaching evolution or only briefly deal with it in classroom settings to avoid controversy, despite the importance of evolution in modern science. Not only is this an omission of a major section of biology, it is a further disservice to students because it allows them to continue to embrace major misconceptions without challenge (Nickels 1996). However, with advocacy groups on both ends of the political spectrum poised to sue any school district that offends their interpretations of the First Amendment, who can blame teachers for shying away from the subject of religion (Thomas 1996)?

Defining the Two Sides in the Conflict: What is All this Fuss About?

There are many facets to the evolution/creationism controversy. For purposes of operational definition, I will characterize the general theory of evolution, the nature of religion, scientific creationism, the nature of science, the educational value of scientific literacy, the establishment of science standards, and official statements from scientific groups about this debate.

Evolution: The various component theories of evolution encompass science fields such as physics, astronomy, geology, biology, paleontology, anthropology, genetics, biochemistry, etc. These theories comprise the formation of the universe, the evolution of Planet Earth, the origins of life, and diversification of life on Earth.

The scientific perspective presented is of such all-embracing scope as to dazzle the mind. Reason and evidence seem to indicate that the present observable universe arose between ten and twenty billion years ago from the expansion and transformation of a primeval, unimaginably condensed mass of fundamental particles. In the nuclear furnaces of very hot stars, the heavy elements of the universe were formed, and through super-novae explosions subsequent generations of stars and planets such as our Earth came into existence. As Earth cooled, simple molecules in surface puddles formed organic chemicals – the building blocks of living systems. From such primeval organic

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soup emerged complex organizations of matter that were self-reproducing. Through genetic mutations, Darwinian natural selection, and the development of symbiotic relationships, sprang the present diversity of living organisms, which climaxed in self-aware human beings that acquired language, society, and culture. The theories involved in biological evolution comprise a logically constructed history of life on Earth as well as an explanation of discrete formation of species and the origin of life. These processes of cosmic, planetary, and biological evolution are viewed as generally continuous from beginning to end, and scientists find no need to invoke any non-natural causes to explain the process. It is an emergent process that grows in richness and diversity according to natural laws (Hill 1998, Zook, 1995).

Bodies of evidence that support evolution are substantial. Since my field is life sciences, I will give examples of evidence supporting biological evolution. They include the existence of natural, discrete, hierarchically nested groups of organisms; the congruent nature of molecular, anatomical and physiological similarities between organisms; both the adaptive and non-adaptive nature of specific biological characteristics of organisms; and a paleontological record showing a combination of temporal succession of species, intermediate or transitional forms, and ecologically coherent fossil assemblages (Nickels 1996, Zook 1995).

The particular portion of the general theory of evolution that seems to most disturb those of religious backgrounds is the biological evolution of humans from lower life forms. Aside from the apparent contradiction with traditional religious beliefs concerning the origin of mankind the inference is frequently made that if the theory of evolution is true, there is no God. This causes unnecessary trepidation. There is nothing in the theory of evolution, molecular biology, or physics that would rule out a purposeful goal for the universe. Though no place has been allotted in the theory of evolution wherein to insert the intervention of an agent such as God, many scientists stand in awe at the amazing mystery that underlies this process. It is upon this mystery that the theologian is called on to offer reflection (Hill 1998).

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Religion: Traditionally, the term religion refers to the various institutions organized to teach their members a way of life that includes an interpretation of reality based on the belief that there is a purpose to the universe and that it is guided by one or more supernatural influences. God, Allah, Jehovah, Buddha, The Hero Twins, Mother Nature, The Great Spirit, and countless others are the names given to such supernatural influence. Religion can also be viewed as a cultural tradition in which children are raised. For the faithful, morality and core values are based upon their religious beliefs.

I subscribe to the more general and somewhat controversial definition; the broadest meaning of the term religion identifies it as any cause, a principle, or an activity pursued with zeal or conscientious devotion. The word itself derives from a Latin term meaning to tie fast and relates to another Latin word meaning to rely. An understanding of religion must take into account its distinctive qualities and patterns as a form of human experience (Encarta 98 Desk Encyclopedia and Webster's 1913 Revised Unabridged Dictionary).

An individual's personal beliefs are intertwined with his or her emotions and sense of security in life. This is as true for the scientist as for the theist. I endorse the view that for individuals and groups of persons science can be a form of religion because it is a way of life – a mental construct that can be used for the interpretation of reality and as the basis for making individual and group decisions. Natural Laws and theories are the doctrines of science and its methodical practices are its rituals.

There are many points of interplay between science and traditional theistic religions. For instance, today's scientific insights have been able to help us be more aware than in past times of the immensity and awesome depth of the universe. As a result, theology has a much deeper and richer notion of reality to reflect upon than it did in biblical or even medieval times (Hill 1998).

As previously pointed out, most Americans are at least nominally theistic and a large percentage of us build our lives on theistic principles. In the instance of the evolution/creationism question, the public school controversy and heated arguments have become a symbol for the deep concerns the faithful have about their children. Many

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religious persons fear that the good life is about over; that society has become self-indulgent and immoral; that science is going too far, forcing wrenching decisions on children who are unprepared to make them (Jackson 1995).

The New Christian Right holds that traditional God-centered Christianity has been replaced in American society and schools by a humanity-centered secularism. They claim that this reliance on man instead of God has caused many of our modern ills, including rising violence, crime, drug use, abortion, divorce, homosexuality, and child abuse and abandonment (Parker 1988).

Many with religious convictions view the general theory of evolution with exasperation and an eye of incredulity. They infer from this theory that scientists proclaim this world to consist only of inanimate matter held together by laws of inertia and gravity, evolving by chance and natural selection, and gradually running out of energy as it moves to a blind end. Here the universe is a merely enormous tract of space and matter, and all life is reduced to chemistry, with no consideration given to ultimate causality or the presence or function of the spiritual (Hill 1998). The purpose of the universe and life as we know it is one of the major pursuits of the religious. Their criticism of evolution and science in general typically revolves around its lack of an identifiable purpose.

Scientific creationism: In essence, scientific creationism is a defensive response of religious scientists to the theory of evolution. Scientific creationism is not the same thing as the religious doctrine of creation. Creation science is a movement started by groups of religious scientists who founded organizations and research facilities, chiefly at private religion-oriented colleges, with the intent to gather evidence and perform experiments using the methods of science to support the religious doctrine of the divine creation of the universe and all known life. The teaching of scientific creationism uses methods of science to critically analyze the theory of evolution, usually casting doubt on its premises and proposing alternative hypotheses that are more in harmony with religious beliefs.

Nature of Science: Many religious devotees suppose that an understanding of religious beliefs and practices would convince the “gentiles” of the errors of their ways.

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In the same sense, many subscribing to the tenants of science make the assumption that if critics of evolution could only understand its theories, they would believe them. Science educators typically feel that the reason there is such controversy over evolution is that lay persons carry misconceptions about the theory of evolution and the general nature of science (Clough, p.409).

So, what is the nature of science? Einstein, quoted by Clough (1994), described science with an easily understood analogy:

“In our endeavor to understand reality we are somewhat like a man trying to understand the mechanism of a closed watch. If he is ingenious he may form some picture of a mechanism which could be responsible for all the things he observes, but he may never be quite sure his picture is the only one which could explain his observations. He will never be able to compare his picture with the real mechanism and he cannot even imagine the possibility or the meaning of such a comparison.”

Four aspects of modern science are critically important to understanding the nature of science:

1. The two most fundamental assumptions of science are (a) the universe operates according to natural laws that do not change over time and (b) through systematic investigation and human reasoning we can identify these laws.
2. It is essential to distinguish between science and non-science – the measurable and immeasurable.
3. Scientific knowledge is fundamentally and irrevocably uncertain.
4. Despite these many sources of fundamental uncertainty, science not only manages to produce ideas that are reliable enough to have immense practical importance, it also manages to make considerable progress in developing major new insights.

Students' misunderstanding of the term “theory” may lie at the roots of scientific misconceptions. A theory is a set of universal statements that explain the natural world. Theories are powerful tools. Scientists seek to develop theories that (1) are internally consistent and compatible with the evidence, (2) are firmly grounded in and based upon

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evidence, (3) have been tested against a diverse range of phenomena, (4) possess broad and demonstrable effectiveness in problem-solving, (5) explain a wide variety of phenomena (National Science Teachers' Association 1997). Many incorrectly view a scientific theory as a speculation, a guess or conjecture (Firenze 1997). However, theories and other scientific explanations change. New theories emerge and other theories are modified or discarded. The theory of evolution is generally accepted by scientists as the probable path of the natural development of the universe and all life to its present state.

Another important point to establish is that creationism is not a conceptual construct that can be tested by science; though, in the reverse, an understanding of nature and man's system of scientific study may lend deeper consideration of reality. Many outspoken science educators go so far as to claim that any credibility given to creationism compounds the already prevalent student misconceptions concerning the nature of science. One such published educator, Michael Clough, makes a statement that seems to embody the opinion of the majority in science professions:

“Most creationists' empirical claims are plain nonsense. Moreover, acceptance of evolutionary theory is largely dependent on an individual's understanding of the nature of science.” (Clough, p.410).

Scientific Literacy: Why is an understanding of the theory of evolution so important to educators? High school biology is currently the last science course taken by the majority of American voters. Cultivating an understanding and appreciation of science as a process that produces ideas that are likely to eventually be made obsolete as they are replaced with enhanced and improved knowledge should be essential to science education. Students, as future voting citizens, need to learn to value the scientific process that generates much of the technology and medical techniques upon which society has come to rely (Nickels 1996).

Scientific disciplines with a historical component such as astronomy, geology, biology, and anthropology, cannot be taught with integrity if evolution is not emphasized (National Science Teachers' Association 1997). Further, every branch of modern biology is rooted in the evolutionary perspective of the organic world. There is no contemporary

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scientific rival to evolutionary theory as the major explanatory paradigm across the discipline. High school educators, regardless of their personal convictions on the evolution/creationism issue, are justifiably concerned about preparing their students to deal with college biology courses. Anne Sinclair and her fellow researchers at Southeastern Louisiana University studied the relationship between college zoology students' beliefs about evolutionary theory and religion. One of their more salient conclusions pointed out:

“Secondary instructors are ineffective in ‘widening the world view’ of their students when evolution is treated superficially in order to avoid controversy with Biblical literalists who often support the teaching of creationism, even to the exclusion of evolutionary theory. As a result, college biology students are unprepared both emotionally and academically to confront the more rigorous curriculum.” (Sinclair, p.118).

Also of concern is the fact that many students devalue science and scientific researchers. Science educators must be diligent in counteracting a negative classroom characterization of anyone who has devoted a great deal of time in scientific work, but who may be regarded as a nonsense theoretician. If one of the purposes of public education in society is to have students become aware of and understand the nature of science and develop scientific habits of mind, a crucial message that must be included in science instruction is that understandings result from the contributions of many individuals, even those who fall into stereotyped ridicule (Zook 1995).

Science Education Standards: Several attempts have been made to politically define scientific literacy. The Florida Department of Education developed the Sunshine State Standards (1998) to address the important concepts and abilities that students ought to cultivate from public education. These state standards are similar to those of nearly every other state in our nation and closely reflect the standards recommended by professional scientific organizations. At the middle school level (6-8) Florida students must learn the following in science classes before entering high school:

Processes of Life Standard 2

“Know that generally organisms in a population live long enough to reproduce because they have survival characteristics.”

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How Living Things Interact with Their Environment Standard 1

“Know that the fossil record provides evidence that changes in the kinds of plants and animals in the environment have been occurring over time.”

“Know that biological adaptations include changes in structures, behaviors, or physiology that enhance reproductive success in a particular environment.”

The 9-12th grade science section of the Sunshine State Standards mandates that by graduation from high school a student must do the following:

Earth and Space Standard 2

“Know various scientific theories on how the universe was formed.”

Processes of Life Standard 2

“Understand the mechanisms of change (e.g., mutation and natural selection) that lead to adaptations in a species and their ability to survive naturally in changing conditions and to increase species diversity.”

Though these standards do not indicate to what depth a teacher must present the general theory of evolution, standardized tests such as the SAT, ACT, FCAT, and the AP exam require a level of understanding that, in part, is achieved through an in depth understanding of this unifying theory of biological science. Though not intended for this purpose, teachers are frequently evaluated on the basis of their students' scores on such standardized exams. Job security seems to require science educators to comply with the state standards.

Professional Science Organization Statements: There is no shortage of informed opinion concerning the controversy surrounding evolution education, creationism, and scientific creationism. At least thirty-two scientific organizations (including the National Science Teachers Association, National Association of Biology Teachers, and the Society for College Science Teachers) have published specific statements advocating the teaching of evolutionary principles as the unifying theme that includes the origin, history, and

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diversity of life on earth. In unity, they firmly insist that evolutionary theory is the “megatheme” upon which understanding of the life sciences must hang (Firenze 1997).

The official position of the National Academy of Sciences (1984) is quite clear regarding the invalidity of scientific creationism and the dogmatic truth of evolution. They compared evolution and creationism to the historic views of the Earth being flat or round. “Both views have a place in our lives – but one belongs in the world as we have come to know it and one belongs in history.” Again, “Teaching creationism is like asking our children to believe on faith, without recourse to time-tested evidence, that the dimensions of the world are the same as those depicted in maps drawn in the days before Columbus set sail with his three small ships, when we *know* from factual observations that they are really quite different.” (NAS, p.5).

In painfully plain language:

“The Academy states unequivocally that the tenets of ‘creation science’ are not supported by scientific evidence, that creationism has no place in a science curriculum at any level, that its proposed teaching would be impossible in any constructive sense for well-informed and conscientious science teachers, and that its teaching would be contrary to the nation’s need for a scientifically literate citizenry and for a large, well-informed pool of scientific and technical personnel.” (NAS, p.7).

Up to this point in the National Academy of Science’s statement, scientific creationism has been the sole target of criticism. They went on, however, to say the following of the general religious belief in creationism:

“The hypothesis of special creation has, over nearly two centuries, been repeatedly and sympathetically considered and rejected on evidential grounds by qualified observers and experimentalists. In the forms given in the first two chapters of Genesis, it is now an invalidated hypothesis.... [Creationism] subordinates evidence to statements based on authority and revelation. Its documentation is almost entirely limited to the special publications of its advocates. And its central hypothesis is not subject to change in light of new data or demonstration of error. Moreover, when the evidence for creationism has been subjected to the tests of the scientific method, it has been found invalid. No body of beliefs that has its origin in doctrinal material rather than scientific observation should be admissible as science in any science course. Incorporating the teaching of such doctrines into science curriculum stifles the development of critical

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thinking patterns in the developing mind and seriously compromises the best interests of public education. This could eventually hamper the advancement of science and technology as students take their places as leaders of future generations.” (NAS, p.7).

I must take personal exception with the far reaching implications and culturally devaluating statements made by the National Academy of Science in the latter half of the preceding quote. Unfortunately similar wordings are used in other official publications. In response to line 12 of the preceding quote, I would ask whether the NAS advocates the suppression of the natural, religiously oriented questions that students ask as a result of teaching evolution? There is no possible way of *effectively* teaching evolution without at least acknowledging the various students' religious backgrounds. To attempt to do so alienates the minds of the students we teachers are crusading to reach. In response to line 14, does this position not limit the possible alternatives that one might explore as avenues of critical thinking? I analyze the validity of alternative philosophical views later. In analysis of the last sentence, I honestly ask of my fellow professionals, does belief in creationism truly make a person of less worth to a scientifically enlightened society? Does such belief invalidate their views of life, their moral station, or the importance of their democratic participation in the guidance of our nation's body politic?

I assert that the National Academy of Science and any true scientist who directly attacks the validity of the religious doctrine of creationism have overstepped their bounds and, collectively, have not come to understand the nature or tenants of science – the same lack of understanding which they show to be woefully deficient in public school students. The belief in divine creation and supernatural guidance of nature is outside the realm of science. No one in their right scientific mind can say that creationism is invalid, only that it is impossible to test using the methods of science. Attempting to disprove creationism in public school classrooms is distasteful, insulting to at least 47% of the American public, and antagonistic to the value of multiculturalism that teachers purport to uphold. At the very furthest, an ethical educator could only state that he or she feels that creationism is improbable.

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I do not here advocate the in-depth, curricular-mandated teaching of religious creationism or scientific creationism in science courses. As a scientist, I acknowledge that science is only one way of interpreting reality. As a theist, I believe that the subject of creationism should not be banned from the science classroom and treated as a taboo topic of discussion.

On the other hand, I agree with the National Science Teachers' Association that, "Policy-makers and administrators should not mandate policies requiring the teaching of creation science, or related concepts such as so-called 'intelligent design,' 'abrupt appearance,' and 'arguments against evolution'" (NSTA p.26). This would place the atheistic teacher in the same uncomfortable position as the science student who is told to reevaluate his or her religious upbringing.

Historic Legal Developments: Battles of the Dogmas

Proponents of science want the freedom to teach what they believe to be one of the most important components of modern scientific thought and, with increasing regularity, they have the support of school administrators. Defenders of religious freedoms do not want their children indoctrinated to an idea that has the potential to undermine their faith in traditional religious beliefs and they typically have the support of the majority of their community. Who is right and who is wrong? Is the issue as easy to decide as favoring one side over another?

At times one of the sides of the evolution/creationism battle challenges the other in court. I will now review the most important of the recent court cases, tracing salient developments in the constitutional position on evolution/creationism education in our society and the principle of academic freedom.

Constitutional Considerations: "Our fathers seem to have been perfectly sincere in their belief that the members of the Church would be more patriotic, and the citizens of the State more religious, by keeping their respective functions entirely separate" (*Lee v.*

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Weisman[†] 1992). This is an ideal principle, but there are complications to be resolved whenever policies and practices of governmental institutions (e.g. public schools) seem to interfere with the sense of integrity and religious freedom of teachers or students.

As mentioned earlier, conflicts between school patrons and public school officials over the evolution/creationism issue typically revolve around interpretations of our constitutional guarantee of religious freedoms. The First Amendment has two religious clauses which protect the individual's religious liberty: (1) The Establishment Clause, "Congress shall make no law respecting an establishment of religion," and (2) the Free Exercise Clause, "...or prohibiting the free exercise thereof." One additional note on the Establishment Clause: any legislation that would involve excessive entanglement between government and religion has been ruled as constitutionally unacceptable in the Supreme Court.

Though there had been earlier victories in the battle for the advancement of scientific view points, evolution was downplayed in biology classrooms and textbooks until the USSR launched Sputnik, in October 1957, and a frightened Congress and the National Science Foundation funded better math and science textbooks which began including principles of evolution (Parker 1988).

In spite of government enthusiasm to increase scientific literacy of American students, many states maintained laws prohibiting the teaching of evolution in science classrooms. In the 1968 *Epperson v. Arkansas* case, the Supreme Court ruled that any law prohibiting the teaching of evolutionary theory violates the Establishment clause of the First Amendment. This was justified because the law "selects from the body of knowledge a particular segment which it proscribes for the sole reason that it is deemed to conflict with a particular religious doctrine; that is, with a particular interpretation of the Book of Genesis by particular religious groups" (Crockerberg 1988).

The Supreme Court also declared in the *Epperson* decision that "the First Amendment does not permit the State to require that teaching and learning must be tailored to the principles or prohibitions of any religious sect or dogma." The strongest

[†] All legal cases mentioned (in Italics) are either referenced in the works cited section as Supreme Court

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statement made was that government "may not aid, foster, or promote one religion or religious theory against another *or even against the militant opposite*." (Italics added). To the relief of theistic science teachers, however, the Justices reminded school officials in the opinion statement that they should accommodate teachers who are conscientiously opposed to teaching certain types of material [e.g. evolution] based on their religious beliefs.

In 1981, a California Superior Court found in *Segraves v. State of California* that the State's Board of Education's Science Framework gave sufficient accommodation to the religious views of Segraves, contrary to his contention that class discussion of evolution prohibited his and his children's free exercise of religion. The anti-dogmatism policy written in the Science Framework provided that class discussions of evolution should emphasize that scientific explanations focus on "how", not "ultimate cause," and that any speculative statements concerning origins of life, both in texts and in classes, should be presented conditionally, not dogmatically (*Segraves v. California*, 1981, Sacramento Superior Court #278978). This ruling is reflective of many court cases across the nation at the state level.

By the mid-1980s, the religious fundamentalists, believers in a literal interpretation of the Bible, had been defeated in their attempts to use state legislation to exclude evolution from the nation's classrooms. In response, several fundamentalist Christian organizations began to promote the idea that the Genesis account is supported by scientific data and referred to this pursuit as "creation science" or "scientific creationism" (Bjorklund 1992). Fundamentalist leaders then adopted a revised strategy, including a new platform used in attempts to establish legislation with the following planks:

1. Evolution is an aspect of a religion, which they call "secular humanism," and so should not be taught in the public schools.
2. The appropriate approach to the study of the origin of man is scientific creationism.
3. If the public schools insist on teaching evolution, they should also include scientific creationism in their curricula (Parker, p.339).

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As a result of fundamentalist activities, several states actually enacted laws requiring balanced treatment of evolution and scientific creationism. Counter suit by the opposition went to the federal level. In *McLean v. Arkansas* in 1982, a federal district court held that a state law, identical to the law later struck down by the Supreme Court in *Aquillard*, requiring that the public schools “shall give balanced treatment to creation-science and evolutionary-science” violated the Establishment Clause. They rejected creation science and concluded, “since creation-science is not science, the conclusion is inescapable that the only real effect of (the act) is the advancement of religion” (*McLean v. Arkansas Board of Education*, 529 F. Supp. 1255, 50 (1982) U.S. Law Week 2412).

In essence, proponents of evolution education won the battle, leaving many of the faithful feeling oppressed and with a sense of injustice. In an attempt to ease tensions, the U.S. Supreme Court decision in the 1984 *Lynch v. Donnelly* case established an important guideline regarding religion/government relationships in education. “Public educators do have a responsibility not to deliberately offend those whose beliefs or nonbeliefs may differ from the majority.” This admonishment provided a principle of consideration in an argument that, too frequently, gets very ugly between students, teachers, and school administrators. It does not satisfy the desires of the religious concerning evolution education but does establish a requirement of common decency. The Constitution, “affirmatively mandates accommodation, not merely tolerance, of all religions, and forbids hostility toward any.”

As a death blow to creation science, in 1987, the United States Supreme Court in *Edwards v. Aquillard* struck down a Louisiana statute that prohibited the teaching of evolutionary theory in the public schools unless “creation science” was also taught. The Court held that the purpose of the act was to advance a particular religious doctrine in violation of the Establishment Clause of the First Amendment, which prohibits laws “respecting an establishment of religion.” In statements given, the Justices referred to the illegal legislature’s purpose as a “sham.” The court said that the law attempted to “discredit evolution by counterbalancing its teaching at every turn with the teaching of

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creation science.” Further they stated, “There can be no legitimate state interest in protecting particular religions from scientific views distasteful to them.”

In a 1992 ruling, the Supreme Court reaffirmed the validity of the set of standards used since 1971 to determine the constitutionality of religion/education related laws. These standards are frequently referred to as the *Lemon Test*, named for an earlier case. They require that school activities (1) be secular in nature, (2) neither advance or inhibit religion, and (3) do not create excessive government entanglement. These standards will undoubtedly be used in all new cases in the future (Peterson 1996).

Academic Freedom of Teachers and Students: The academic freedom issue has its birthplace in Illinois in 1987. A teacher was prohibited by a local school board from teaching creation science in addition to evolution. He filed suit contending that the discussion of religious issues in his classroom was only to encourage students to develop open minds and explore alternative points of view. The federal district court ruled in favor of the school board, citing *Edwards v. Aguillard* (Bjorklun 1992).

Following this case, another plaintiff filed suit. A student claimed that he had a First Amendment right to receive information about creation science and that prohibiting the teaching of creation science interfered with that right. Once again, this suit was dismissed. The justification given was that allowing creation science instruction would violate the First Amendment rights of the other students (Bjorklun 1992).

Supreme Court decisions affirm that, whether viewed as a form of religion or not, the legal suppression of scientific evolution dogmas would stifle educators' freedom of expression and liberty to teach about the scientific way of interpreting the universe. Does this not apply in the reverse as well? Could this principle not be properly applied to the case of a teacher who indicates doubt in the theory of evolution, whether his view is religious or personal?

What if a science instructor teaches the prevailing theories of biological and pre-biological evolution, but criticizes them on the basis of his own opinion that a combination of chance events and chemical laws is probably not capable of creating life, and that Darwinian natural selection is probably not capable of crafting complex organs

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like eyes and brains (Johnson 1995)? Such a professor would surely be accused by some students or colleagues of teaching “creationism,” and thus of improperly inserting a different subject, religion, into what was supposed to be a science course. The teacher may reply that he is teaching only science and bringing critical scrutiny to *propositions that are usually taught dogmatically*. Can a science educator teach critical analysis of the theory of evolution and admit that the existence of a supernatural creator cannot be dismissed as a possibility? Is the professor doing something unprofessional, or even unconstitutional?

In *Lamb's Chapel v. Center Moriches School District* (1993) the Supreme Court unanimously ruled that the exclusion of a viewpoint for no reason “other than the fact that the presentation would have been from a religious perspective” is to violate the constitutional principle that the government may not deny a speaker access to a public forum “solely to suppress the view he espouses on an otherwise includible subject.” In other words, the suppression of a viewpoint merely because it is “religious” is to engage in unconstitutional viewpoint discrimination. However, the Supreme Court did not go so far as to indicate whether this can specifically apply to a public school teacher.

In 1990, the Seventh Circuit Court of Appeals ruled in *Webster v. New Lenox School District* that a school district may prohibit a teacher from teaching creation science. The court held that the school district had not violated Webster's free speech rights when it fired him for voluntarily teaching creation science along with evolution in a science classroom (*Webster v. New Lenox School District*, 917 F.2d 1004, 7th Cir. 1990).

In *Bishop v. Aronov* (1991), a federal appeals court upheld a University's decision not to allow a science professor to express his doubts about the Darwinian theory. The opinion hinted that the University might have been constitutionally obligated to take this action under the first amendment's establishment clause, saying that “the creation/evolution aspect of his lecture could have lent itself to an analysis as found in *Edwards v. Aguillard* (*Bishop v. Aronov*, 926 F.2d 1066, 11th Cir. 1991, cert. Denied, 112 S.Ct. 3026, 1992). This decision also held that University of Alabama administrators

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were justified in forbidding the same professor to tell students that he was a Christian, even though the university allowed professors generally to tell students about their political or philosophical orientation. He was not allowed to express his doubts about the Darwinian theory, even in voluntary, after-hours, private discussions. The court explained that “Just as women students would find no comfort in an openly sexist instructor, an Islamic or Jewish student will not likely savor the Christian bias that Dr. Bishop professes.” This decision seems to imply that an educator may tell students that he or she is a feminist, or gay, or an atheist, and even base arguments or class assignments on that perspective (as many do), but may not say that he or she is a Christian, which the opinion seems to categorize as a species of bigotry (Johnson 1995).

Again, in *Peloza v. Capistrano School District* (1994), the Ninth Circuit Court of Appeals upheld a district court finding that a teacher's First Amendment right to free exercise of religion is not violated by a school district's requirement that evolution be taught in biology classes, regardless of his or her religious conviction. Rejecting plaintiff Peloza's claim that “evolutionism” constituted a form of religion, the Court found that the district had simply and appropriately required a science teacher to teach a scientific theory in biology class (*Peloza v. Capistrano Unified School District*, 1994, 917 F. 2d 1004). The court also upheld the 1991 decision that it is not a violation of the constitutional right to freedom of religion for local school districts to forbid teachers to discuss religious beliefs even outside the classroom, on their own time, and in private with students who voluntarily seek such discussion.

The *Edwards v. Aguillard* holding only declares it unconstitutional to mandate the inclusion of scientific creationism, it does not legally prevent an individual teacher from challenging the theory of evolution as a matter of pedagogical choice. In many cases, however, school officials feel that they must overrule a teacher's decision to explore religious subject materials for fear of violating the law. Misinterpretation of court rulings has inhibited many public school educators or students from studying about the historical significance of religion to the development of modern thinking (Peterson 1996). As shown previously, in these situations, the lower courts have upheld administrators'

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authority to guide local curriculum. This matter has yet to be brought before the Supreme Court for a ruling on the constitutionality of prohibiting science instructors from voluntarily teaching scientific creationism and/or the general religious belief of creationism in addition to evolution. For a more concise summary of court rulings, see Appendix A.

In evolution versus creationism, there are two constitutional principles in tension with each other: one forbids viewpoint discrimination and the other forbids the presentation of religious viewpoints in some contexts – even as a corrective to balance the presentation of naturalistic viewpoints. The irony in this situation is that there is no definition of “religion” for legal purposes. The Supreme Court has said on occasions that a religion does not necessarily involve a belief in a supernatural creator, so that there seems to be no principled way of determining whether such belief systems as Marxism, science, and feminism are or are not religions. As a side note of curiosity, court rulings excluding religious speech from a public forum almost always involve Christianity (Johnson p.17).

While the debate over creation science and evolution is not, properly constructed, a scientific debate at all, it does raise serious questions about the proper treatment of religious as well as non-religious believers in the public schools (Crockenberg 1988).

By the same reasoning used in justifying the 1968 *Eperson* decision could not it be asserted that legally prohibiting the voluntary teaching of creationism and mandating that evolution be taught (regardless of the religious convictions of the teacher) are other forms of selecting from the body of knowledge a particular segment to be legally proscribed? Can we preserve the establishment clause of the First Amendment without unduly restricting the free exercise clause?

If we say that it is important to society that students learn the scientific method or logical reasoning, even if it tends to undermine their religious beliefs, and at the same time insist that the establishment clause excludes religion from publicly funded education are we in the position of silencing an “intellectual heresy?” Education officials then bear resemblance to some religious fundamentalists when it comes to intolerance – a practice

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viewed as repugnant by liberal thinkers. Are we, in the terms of reformed epistemology, rooting educational practice and policy in one properly basic belief and excluding another (Lebman 1989, Milligan 1996)?

One new development may become a cause for future controversy. In 1996, the first-ever presidential directive on religion and public education was released. Building on a legal summary developed by dozens of religious and civil-liberties groups, the directive summarized the new legal consensus. It was sent to every superintendent in the United States with orders from the president that schools should not be “religion-free zones” (Thomas, p.65). This mandate is vague, at best. Since the President is the one who appoints Supreme Court Justices, what might this new policy portend?

Cognitive Development Questions: Milk before Meat

When considering education psychology and developmental theories, one particular researcher comes to the forefront in addressing the question of how and when to teach evolution and creationism. Piaget postulated that children do not typically gain cognitive readiness for abstract reasoning until late adolescence. In fact, more recent studies indicate that many adults never surpass the concrete functioning stage of mental processing (Glynn 1991). In terms of teaching the admittedly abstract general theory of evolution, perhaps educators should use a “milk before meat approach.” Student difficulties with evolution may lie in their inability to understand abstract concepts of genetics and how they relate to natural selection (Firenze 1997). Also, many religious K-12 students may not be ready to form alternate mental constructs sufficiently abstract to understand the significance of the finer points of evolutionary theory or its integrating relationship to all fields of science. Lawson and Worsnop (1992) studied this question. Their research data suggested that level of cognitive development does indeed affect the ability of students to understand or even reconcile differences in belief systems such as evolution and creationism.

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With regard to the intellectual controversy at hand, one could ask, "Is truth democratic?" No! Of course not. But, does society have the right to determine at what stage children should be exposed to alternate propositions of truth? Yes!

Concepts and theories that, through long familiarity, seem satisfying to a scientist or science teacher may not be so easily credible to his or her students. How tenable to a concrete thinker is the idea that a protozoan ever changed, even over a long period of time, to become an elephant? How believable is the idea that bacteria seen in the microscope might have been an ancient forerunner to the human species (Jackson 1995)? Perhaps education officials in charge of state and local standards should take this into consideration in the timing of the introduction of the theory of evolution to science students. Further studies need to be done to determine when, in development, a student is ready to tackle the abstract challenges of evolution theory.

A whole separate issue here is the fact that understanding does not automatically equate to belief. While it is true that belief in evolution does not necessarily preclude a belief in God, a good understanding of evolution does not necessarily lead to acceptance of the theory (Dagher 1997). This is the case not only for the student but also for the scientist.

The National Association of Biology Teachers acknowledges that students can maintain their religious beliefs and learn the scientific foundations of evolution (NABT 1996). The actual problem is not at the level of understanding but at the level of accepting this understanding as true (Dagher 1997). In a study done by Bishop and Anderson (1990) students were taught and assessed until a satisfactory mastery of evolutionary concepts was attained. They found that a demonstrably greater understanding of Darwinian theory of the mechanisms of evolution did not translate into a significant change in students' basic beliefs.

When dealing with these research results, Jackson asked, "What makes a scientific understanding of evolution so difficult to achieve" (Jackson, p.588)? His personal belief is like that of the scores of science educators who have written on this topic: the notion that if you understand the theory you will automatically accept it as your

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new dogma of life. Though I am a scientist, I emphatically assert that a logical understanding does not equate to acceptance – and it doesn't have to. The idea that the student does not understand the theory of evolution until he or she believes in it is one of the main reasons that scientists and lay persons argue about evolution and creationism. The mere use of this line of reasoning implies that science educators are on a missionary crusade at least as aggressive as that of traditional religions. Most science educators seem to be out to convert the religious heathens to their form of doctrine. And, as established earlier, those who defend their personal convictions of faith in public schools currently stand on increasingly shaky legal grounds from which to seek recourse.

Alternative Views: Is There One Right Way of Knowing?

Alternate ways of knowing: With regard to alternate ways of knowing, two prominent analytical constructs exist: orientation and world-view (Dagher 1997). An orientation means a tendency for an individual to understand and experience the world through a particular interpretive framework, embodying a coherent set of beliefs and values. A world-view is a culturally dependent, implicit, fundamental organization of the mind. This implicit organization is composed of presuppositions or assumptions that predispose one to feel, think, and act in predictable patterns. Both constructs are intellectual tools used to analyze possible alternate ways of knowing.

Many of the criticisms raised by both sides of the evolution/creationism issue amount to little more than finger pointing and petty bickering. The fundamental question here is whether a person can choose an emotion orientation over reason and yet be considered an intelligent contributing member of society (Grobman, p.337). Is it essential to the future of humanity that today's students be indoctrinated to one particular cognitive orientation (science) that is inherently based upon assumptions, inferences and evidence, but (like all alternative views) not absolute proof (Kessler 1993)?

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During the preparation of this paper, I found that the majority of scientific writers and science organizations seem to be of the opinion that a person is scientifically illiterate if they disbelieve any one of evolution's carefully constructed and thoroughly logical theories. Can a person be literate and even conversant about science and yet legitimately choose another way of life?

Concerning religion as a way of living, the National Academy of Science has stated, "It is false to think that the theory of evolution represents an irreconcilable conflict between religion and science. A great many religious leaders and scientists accept evolution on scientific grounds without relinquishing their belief in religious principles" (National Academy of Science, p.6). This, in no way, means that one *must* reconcile the two very different systems of belief into a hybrid cognitive construct in order to be "intelligent."

Science, though powerful, is only one way of thinking. One can think with the heart and still make sound decisions. Yet, state education departments seem to disagree. The way state science education standards are organized and set forth at present seems to assume that everyone will become essentially an academic person. "The popular assertion that all children can learn probably has some truth in it, but not if it is assumed that all will learn in a strictly academic mode" (Clinchy p.88).

A fairly recent addition to the pool of educational theories that gained wide acceptance and has taken swift hold of many educators' approaches to teaching is Howard Gardner's theory of multiple intelligences. He postulates the existence and equal validity of the following seven ways of knowing. He called them forms of intelligence.

- | | |
|-------------------------|---------------------------------|
| 1. Verbal/Linguistic | 5. Musical/Rhythmic |
| 2. Logical/Mathematical | 6. Interpersonal |
| 3. Visual/Spatial | 7. Intrapersonal (Lazear 1991). |
| 4. Body/Kinesthetic | |

In light of such a theory, it is impossible to justifiably say that the logical/mathematical (scientific) orientation of intelligence is more valuable than all others to the success of our society's future.

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Is science a religion? Science is a way of interpreting reality. It has its own collection of asserted truisms. It has a formal way of initiating novices to its practices and methodologies. It acknowledges mysteries and unknowns, yet attributes none of them to a deity. The Supreme Court has said on occasions that a religion does not necessarily involve a belief in a supernatural creator (Johnson, p.17). Again, using the most liberal meaning of the term, science is a religion – a way of life that includes methodical observations of reality, logical conclusions based on empirical evidence, and reasonable decisions based on hard, physical data. Scientists trust in human reason. Religious critics have named exclusive faith in self “secular humanism.” With science, man is his own god and his intellectual constructs become the object of devotion and worship. Darwinian evolution is the great sacred cow of academia because it provides essential support for the naturalistic philosophies that currently dominates university and political life (Johnson 1995).

Scientific creationism is not science: The underlying assumption of science is that the universe operates through natural, identifiable laws that are immutable. The underlying assumption of most deity-centered religions is that the natural laws of the universe, as humans perceive them, are not necessarily inalterable and that there are operations at work in the universe that we cannot detect with mundane senses or even through technological ingenuity.

Religious assertions are not testable through methods of science. There are many areas of science that may serve to strengthen beliefs of the religious, but the essence of religious belief is founded on non-empirical evidence.

This does not mean that discussion of religion is unwarranted in science classrooms. Religion has played an historical role in the development of our society. Biology classes have a heavy history component, tracing the growth of scientific thought. It would not be inappropriate to at least point out that not everyone currently believes in evolution. The words “creationism” and “God” should not be treated as taboo. I do agree, however, that in depth analysis of religious beliefs and exhaustive cross-

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examinations of science and theological philosophy have no place in *public school* science classes.

What is faith? Dr. Jeffry Milligan (1996), professor and dean of the Division of Arts and Sciences at Murray State College, asserts that faith is not knowledge. In *The Republic*, Plato defines knowledge as “direct acquaintance” with the thing known (Hoitenga p.7-8). This definition of knowledge, however, can only apply to a small portion of what we know, even in science. For example, one can know that the world is round without ever personally seeing it from space or by circumnavigation. This fact is not known by “direct acquaintance” but by the word of an authority figure. How is this different from the religious believer’s “knowledge” of the existence of God? The difference lies in the possibility of confirming the belief. All astronauts have been directly acquainted with the fact that the Earth is not flat. In the case of science theories that cannot be verified by direct acquaintance, such as evolution, how is scientific belief justified as knowledge? Through human reason.

Faith is belief in a proposition – which may or may not be true – that cannot yet be justified through science. Faith, unlike science, rests on trusting something or someone other than self. “Wherever there is an exploratory openness to the new, together with a humble abandonment of the will to subject the universe to the contours of our own limited intellects, there is faith” (Hill, p.388).

Belief remains faith as long as it is not or cannot be confirmed through direct acquaintance or justified logically. If this belief is true and it is confirmed in either of these ways, it ceases to be faith and becomes knowledge (Milligan 1996). To one of religious conviction, however, faith may become a sure knowledge as a result of a lifetime of gathered personal evidence. Many of the faithful declare solemnly that they know God through direct acquaintance or through spiritual experiences that serve them in the same manner that empirical evidence serves the scientist. Spiritual knowledge diverges from science in its inability to directly communicate its evidence or chart spiritual data points. With spiritual knowledge of the existence of God, faith as a mixture

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of belief and trust as well as the believer's relationship to scripture can be rational in a way quite different from but no less valid than the rationality of science (Milligan 1996).

Though many post-Enlightenment philosophers and social critics – including Dewey – have predicted the demise of religious “superstition” and “dogma” in the face of advancing scientific knowledge and secularization, religion has refused to die. In fact, contemporary studies of religion in the United States demonstrate that the percentage of Americans professing religious beliefs is as high now as it has ever been and that, contrary to expectations, it is precisely the more “fundamental” denominations that have shown the most dramatic growth (Milligan 1996).

What is the interplay between cognition and affect (faith)? Several researchers in science education have tried to analyze how patterns of thought and faith affect one another. Educational researcher, Zoubeida Dagher, of the College of Education at the University of Delaware, performed an insightful study (1997) exploring how university biology majors accommodate the theory of biological evolution with their existing religious beliefs. In analysis of the collected data, students typically developed one of four major orientations or world-views. They (a) accepted evolutionary ideas basing themselves in scientific arguments; (b) did not accept evolutionary ideas, presenting arguments from a religious or another antievolution perspective; (c) reinterpreted the theory, assimilating religious and scientific concepts from a compromise perspective; or (d) stayed neutral, reflecting either a non-committed or a confused perspective.

In order to reconcile religious dogma and science, Dewey suggested that one must reinterpret religious truths as having symbolic rather than literal meaning and to recognize that such truths may be found in other than the traditional religious sources. Such reconciliation is an intellectual evolution from “literalism and dogmatism” to the “denial of a personal God, immortality of the soul, and a historical Jesus” brought on by the challenges of modern science and reason (Milligan p.73). As Dagher demonstrated, some students indeed assimilate religious and scientific ideas in this way. The majority of Americans, however, find this recourse a contemptible corruption of convictions or personally dissatisfactory at the least (see Gallup polls already listed).

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Whether or not secular, scientifically oriented education is in fact antagonistic to students' religious belief is the question (Milligan 1996). I like to use a non-conventional, secular metaphor to describe the interaction of cognition and faith. If the sides to the evolution/creationism controversy were characters in an X-files episode, science would be the always-critical Doctor Dana Scully and religion would be the open-minded FBI Agent Fox Mulder. Each seems to moderate the other's extremes. Though an effective team, science and religion find themselves at odds, yet seem inexorably codependent. The truth is out there and society seems to always be on the verge of finding it out.

Scientific indoctrination: Many scientists take offense at anyone suggesting a critical analysis of evolution as "just a theory." Many are not satisfied unless it is taught and taken in as pure, proven, and irrevocable truth. When the issue is boiled down to the essentials, both evolution and intelligent design theories are just exactly that – comprehensive theories put forth by humans to describe and explain our perceptions of life and our place within reality.

Jackson (1995) points out that teaching the basic notions of evolution might have the greatest influence on young minds if introduced early in their intellectual development, when they are least likely to be able to fully understand and appreciate them. But, attempts to teach younger students about evolution could easily fall into the trap of long-term authoritarian indoctrination, one of the very tactics that many scientists and educators find so distasteful about the educational practices of many organized religions.

Eugenie Scott (1996), Director of the National Center for Science Education, asked, "what makes well-meaning people fight so hard to keep children from learning a basic scientific principle" (Firenze p.12)? I would respond by asking, "what makes well-educated people fight so hard to keep children from learning or maintaining a basic spiritual principle?" If the argument is true that religion indoctrinates children who do not yet have the capacity to critically consider dogmatic teachings, perhaps Dagher's (1997) observations are critically important. He pointed out that many science students

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“feel justified in accepting the theory of evolution dogmatically on the basis that, since it is stated in the science book, it must be true” (Dagher p.439).

This perceived pressure to accept the theory of evolution, indirectly implied or openly applied, justifiably disturbs students and parents from strong religious backgrounds. In our nation, public schools are commonly viewed as the authority over education. We often forget that parents, not schools, have the primary right to control the religious and educational upbringing of their children. Parents have every right to be concerned about whether the influences of the public school might negate the teachings of home and church (Thomas 1996).

The point here is that thinking and analysis require time, effort, and careful maintenance of intellectual freedom. Ability to fruitfully consider the arguments of both sides requires open-mindedness and the availability of accurate representation of both sides of the issue. Students must be permitted to make up their minds as they see fit. In a failing school system and in a society of failing homes and values, how often are these idealized conditions met? The path of least resistance to achieving liberal definitions of scientific literacy is scientific indoctrination. This, however, is the exact opposite of intellectual development that is needed in our society and, therefore, must be avoided.

Treatment of the issue as one of multiculturalism: There is a new and growing drive in the education profession to reinvent curriculum with a multicultural approach. The underlying ideal is that no one culture can be valued above another and that all student backgrounds must be honored in order to educate students in a spirit of inclusion and respect (Levine p.343-348). This relates to the subject in that, if viewed as a general form of religion, those subscribing to scientific thought, research, and decision-making can be said to belong to their own culture. Jackson et al. concluded the following from their study of evolution educators:

“We view conflicts between religion and science regarding evolution as a bona fide example of a multicultural issue in education. These theoretical perspectives are inconsistent with the common tendency among science professionals to view or treat orthodox Christian students in a manner unconscionable with others – to disrespect their intellect or belittle their motivations, to offer judgments based on stereotypes and prejudices, to

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ignore threats to personal self-esteem, or to deny the de facto connection of some scientific conceptions to the morals, attitudes, and values of individuals with such religious commitments” (Jackson 1995).

In terms of multiple intelligences, arguments over which orientation or world-view is the most correct will do no good for anyone. Instead, it is wisest to treat the evolution/creation issue as multicultural.

Pedagogical Suggestions from the Experts

Opinions about evolution education abound and are varied. I will summarize a few of what, in my opinion, are the most promising and objective ideas proposed by education researchers.

Instruction should not seek to change the student's minds: Sinclair (1997) contended that the intent of instruction should not be to change the students' minds, but should be to present the evidences that support evolutionary theory in such a way as to invite students to safely explore the alternatives. This allows them to utilize critical analysis (a skill science teachers aim to develop in their students) as they reflect and weigh the strengths and weaknesses of the various scientific and religious arguments.

Teachers must not advocate religions nor deny the possibility of deity: To reiterate the stance of the National Science Teacher's Association (1997), because science can only use natural explanations and not supernatural ones, science teachers should never advocate any religious view about creation, nor advocate the converse: that there is no possibility of supernatural influence in bringing about the universe, as we know it. Teachers should be nonjudgmental about the personal beliefs of students. This does not imply that the subject of creationism should not be mentioned at all but rather, no single religious belief should be championed in the science classroom.

Allow students to discuss their beliefs: Sinclair (1997) and Dagher (1997) both maintain that students need to discuss their beliefs about creation origins and evolution before and throughout instruction and give reasons why each of the arguments is

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compelling. The purpose of such discussions should be to clarify concepts, beliefs, and conflicts without pressures to convert any party to any one belief about science or religion. This enhances participation and during the process students often come to realize that they are not required to make an “either/or” choice between scientific theories and religion.

Allow for exploration of alternative beliefs: Crockenberg (1988) made the point that, ultimately, school people will have to learn to include a multicultural curriculum approach that includes the topic of religion. In *Engle v. Vitale* (1962) the Supreme Court stated, “It might be said that one’s education is not complete without a study of comparative religion and its relationship to the advancement of civilization.” Students need to have a cursory understanding about the varieties of contemporary religious belief in the United States and the constitutional right of each of us to believe or not believe as we so choose, free from unwarranted interference by the state. Learning about the religious beliefs of others, and learning to tolerate, or even celebrate, the rich diversity of those beliefs, is at least as important in contemporary America as learning about evolutionary theory.

Don’t force students to choose: Clough (1994) and Firenze (1997) indicated that if students are faced with a choice between evolutionary theory and their personal religious conviction, science will most certainly lose. When teaching the scientific basis for evolution we must not design our classes to make believers lose their faith but, rather, encourage them only to look carefully at the difference between science and non-science.

Stress understanding and the nature of science rather than acceptance of evolution: Clough (1994) and Nickels (1996) agree that students are much more likely to consider evolution if a functional understanding of the theory is stressed rather than acceptance. Not only would students be more likely to consider evolution, but also they would learn a great deal about the nature of science. They will learn that science is a process of determining what to accept in the absence of certainty. This is a life skill that is desirable independent of scientific literacy. Also, emphasizing the tentative nature of all science reduces the direct conflict between evolution and religious dogma. By making

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a clear separation of science from other ways of knowing and by teaching students to think, analyze and compare available information, evolution need not be a controversial subject in science education.

Use the conceptual change model: Clough (1994) and Jackson (1995) both advocate the use of the conceptual change model in teaching evolution. In this theoretical framework, school learning is viewed as the process of interaction of a student's prior knowledge and past experiences with ideas presented and experiences gained in the current classroom. This pedagogical strategy begins with posing questions or placing students in situations in which they see that a new cognitive construct must be formed or a previous one must be adapted to assimilate new understandings. What science teachers say and the manner in which they speak are especially critical. A fundamental tenet of this constructivistic learning theory is that students' views, whether they be alternate conceptions or misconceptions, must be treated with great respect.

School must reflect the community values: Lebman (1989) puts it best: "The local school serves best as it attempts to reflect the positive values of the particular society it serves" (Lebman, p.276).

Conclusion: I Think, Therefore I Have My Own Opinion.

My stand on evolution as a scientist and as a religious believer: As you may have gathered by now, I am a theistic science educator. I look upon the methods and achievements of science with wonder. I value human intelligence and ingenuity, yet I believe firmly in God. I have both scientific and religious doubts about the theory of evolution.

I am convinced that the theory of evolution is one of the most comprehensive frameworks of logic and reasoning ever conceived by human intellect. Such constructs have become powerful tools of scientific exploration. I am not convinced that the theory is true. I do, however, subscribe vehemently to principles of academic and intellectual

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freedom. I see nothing immoral or unethical in allowing children to explore alternative ways of viewing reality. They must enjoy the same freedom to choose what to believe and how to guide their own lives that we enjoy as adults. But when are they ready for such self-determination? The state mandates seem to imply that they are ready by middle and high school. I have seen many exceptions to this. Yet, my place as a science teacher is to follow the general guidelines, trusting that they reflect the collective will of the people.

My most compelling reason to look upon evolutionary principles critically is that I have had many spiritual experiences that conflict directly with all the reasoning of scientists. I feel that I have come to know of God's existence through direct acquaintance and a lifetime of gathered personal evidence. I cannot account for all the evidence and arguments of science to the contrary; I'm not intelligent enough to do so. As for the origin of man, my religious belief prevents me from accepting the principle of evolution but this does not mean that its theories are illogical. It means that in this particular aspect my faith is stronger than my reason. As a scientist, I agree with other critics of evolution (e.g. Johnson 1995) that new mechanisms must be discovered to account for such phenomenon as the evolution of fundamental biological novelties and complex body plans.

I personally believe that we should not substitute creationism for evolution in schools, even in communities who favor a religious explanation of natural history. We also should not teach evolution to children who are taught to believe in creationism at home without acknowledging and addressing the conflict that they and their families experience as they negotiate these opposing belief systems.

How I will teach evolution: In teaching evolution, I want my students to understand the public controversy over creation and evolution. I have decided that the evolution/creationism debate is a good opportunity to teach students, through the philosophy of science, that science is a very effective, but not the only effective, way of knowing. I see my proper role, as providing relevant information to facilitate and

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encourage students to judge the validity of theories, including evolution, for themselves – after all, this is a basic science skill.

I have chosen to teach evolution using techniques of concept mapping in combination with the principles listed above. In justifying my approach, I argue that there is now more recorded knowledge than can be learned by a student in several lifetimes. One of my goals is to teach students how to get the big picture – derive meaning from concepts that they learn, developing a mental framework for conceptual understanding of science theories. This is more important to me than forcing students to memorize meaningless details.

Tragically, by grade four or five, most students have learned to accept rote learning and prefer it to other more meaningful learning methods (Novak, p.45). Ideas that are novel, powerful, and profound are very difficult for students to think about. They need some *mediating activity* to help them, one that will help students to learn science concepts meaningfully. Well-known educational researcher, Dr. Joseph Novak at Cornell University champions the use of concept mapping because it forces students to learn concepts as organized networks of related information, not as random lists of unrelated facts.

Concept mapping is an effective mediating activity – giving students a visual means to guide their linking of concepts and meanings. To make a concept map one gathers a list of important concepts that are all related. Then the learner draws them out on paper, placing them strategically in relation to one another and drawing linking lines between them to show the direction of perceived relationships. Along with the lines, linking phrases must be included to indicate what relationships exist between the various concepts.

I have successfully used this pedagogical approach in teaching evolution and plan to continue to do so in the future. The concept map included in Appendix B is one I have used as an instructor to organize and present topics for discussion. I also used it as a means to choose relevant details to have the students consider and learn for assessment.

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As this was typically students' first exposure to concept mapping, I chose to have a concrete starting point, "Origin of Life." I distilled down all information at my disposal into three major beliefs that have been popularly held at some point in time with regards to the origin of life – creationism, spontaneous generation, and the general theory of evolution. I very briefly described the first two beliefs, explaining the impossibility of analyzing creationism scientifically and then illustrating the historical scientific discounting of spontaneous generation. I introduced the general theory of evolution as one of the unifying *theoretical principles* in biological sciences, not as an unquestionable fact. Also, I explained that creationism and evolution do not have to be defined as oppositional to one another.

Since the assigned text dealt primarily with evolution as being the answer to the question of life's origins and diversification, I spent (by far) most of my time exposing students to the history and development of this theory. I wanted the students to see how scientists have gone about considering and studying the question of mankind's origins and that of life itself. I felt that the most productive way to do this was to let the students discuss and analyze the sub-theories, evidences, and their own doubts about evolution. The many concepts involved within the theory of evolution make it difficult for students to consider it as a whole without organizing them. Concept mapping proved to be the ideal tool for helping students to get the big picture.

I used concept mapping in three ways in the classroom: (1.) to model how I made meaning from the various concepts dealt with in this topic, (2.) as a review of the material before the exam, and (3.) as an actual assessment question on the exam. I was impressed with how easy this made the presentation of new material. I simply added parts to the map as we progressed through topics. Of key usefulness to me, analyzing student attempts to map out the concepts involved with the origin of life and evolution allowed me to very quickly visualize what preconceptions and misconceptions any student might have had.

Many students did very well on the assessment with the basic elements of structuring concepts in their maps. There were variations in the relationships drawn and

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organization of content. A few forgot key concepts to be placed on lower levels of their maps, but the majority seemed to have achieved the unit objectives I had established. These objectives included: (1) to capture the big picture – demonstrate an understanding of the theory of evolution, (2) realize that there are alternate ways to analyze and believe with regards to the origin of life, and (3) successfully demonstrate the ability to organize knowledge. Genuine understanding of the theory of evolution was reflected in many of the students' maps. Though not the intent of the exercise, some commented that they had been able to better determine what they believe, one way or another, through this activity. A few expressed that this allowed them to justify both creationism and evolution in one. In any case, I was ecstatic that the majority of my students had seriously contemplated the concepts presented in our lessons.

When concept mapping is used properly, educational “subject matter ceases to be a mass of definitions to be memorized or problems to be solved by the routine of plugging-in of numbers or symbols into abstract formulas” (Novak, pp. 48). These techniques work as well with all other science concepts as they do in teaching evolution. I like to use the phrase, “big picture,” when describing my educational philosophy. Like a jigsaw puzzle, I want my students to take the bits and pieces of the whole and learn how to put them together on their own. They may not remember every detail of the puzzle but the skills required to put together the “big picture” will serve them for the rest of their lives.

Concluding statement: As with any academic idea, an understanding about the theory of evolution cannot be injected into the minds of students. “Teachers do not cause learning; learners do” (Novak, p.53). Carefully choosing a pedagogical approach to teaching evolution that maintains academic standards yet respects the intellectual freedom of the student is the key to dealing with the evolution/creationism controversy. It is critical to provide students with a place to stand between the two extremes of this issue, especially if we are to promote an understanding of the nature of science and why scientists feel that the theory of evolution is such a major cornerstone of the biological sciences. Failure to provide such a position of safety to students leaves them far too often

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with an uncomfortable choice (Sinclair, p.124). To achieve a positive environment, in which students can effectively learn the theory of evolution, tolerance must be established in two ways: faithful of the scientific and the scientific of the faithful.

We can ask, "Which is the correct path to finding truth?" Constitutionally, we are free to pursue this question with all diligence. With regard to evolution and creationism in science classrooms, I have contended that intelligent and civil discourse is not only possible but fruitful as well. As for personal belief: to each his own and curses upon us all if we persecute each other because of our differences.

Appendix A:

Summary of Court Rulings

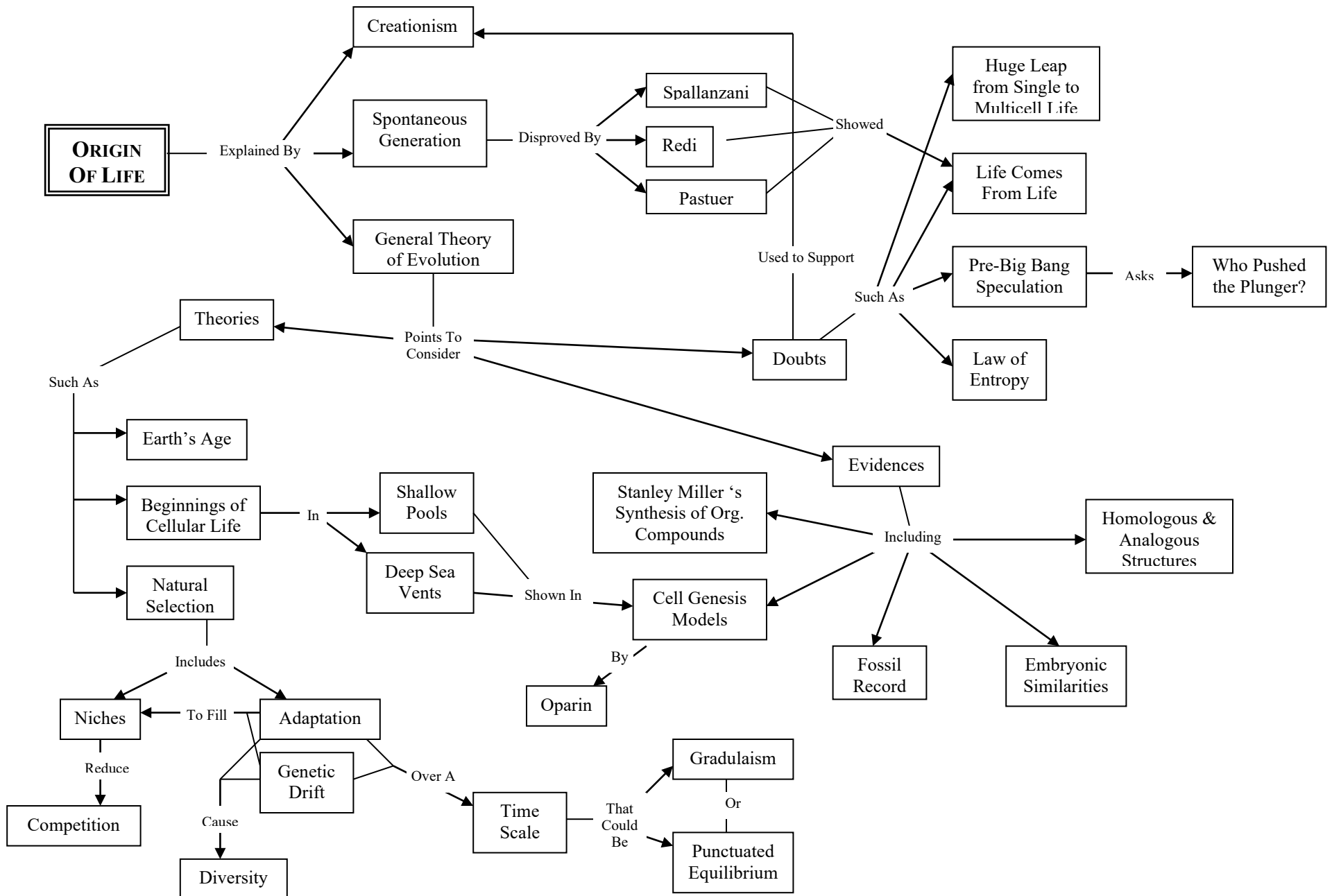
Court Cases:

Rulings:

<i>Eperson v Arkansa</i> 1968.....	No laws may prohibit the teaching of evolution.
<i>Peloza v. Capistrano School District</i> 1994.....	School administrators are permitted to require science teachers to teach the theory of evolution, regardless of their religious convictions.
<i>Bishop v. Aronov</i> 1991.....	School administrators are permitted to forbid teachers from expressing doubts about the theory of evolution.
<i>Edwards v. Aquillard</i> 1987.....	No laws may mandate balanced treatment of evolution and scientific creationism.
<i>Webster v. New Lenox School District</i> 1990.....	School administrators are permitted to forbid a teacher from voluntarily teaching creation science in a science classroom.
<i>Segraves v. State of California</i> 1981.....	Teaching evolution in public schools does not violate student's free exercise of religion.
<i>Webster v. New Lenox School District</i> 1990.....	School administrators are permitted to forbid a student to study creation science in a science course environment.
<i>Bishop (1991) and Peloza (1994)</i>	School administrators are permitted to forbid teachers from admitting to being Christian or discussing religion with students on their own time.

Appendix B:

The Origin of Life Question



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